



## Spring Boot-Hibernate Example

### demo1-boot-hibernate: Files required

|                           |                           |
|---------------------------|---------------------------|
| 1. MyBootApplication.java | 2. JLCAppConfig.java      |
| 3. Customer.java          | 4. CustomerDAO.java       |
| 5. CustomerDAOImpl.java   | 6. application.properties |
| 7. pom.xml                |                           |

#### 1) MyBootApplication.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.CommandLineRunner;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication implements CommandLineRunner {

    @Autowired
    CustomerDAO custDAO;

    public static void main(String[] args) {
        SpringApplication.run(MyBootApplication.class, args);
    }

    @Override
    public void run(String... args) throws Exception {

        // 1.Add the Customer
        Customer cust = new Customer("sri", "sri@jlc", 55555, "Blore");
        custDAO.addCustomer(cust);

        // 2.get All Customers
        List<Customer> mylist1 = custDAO.getAllCustomers();
        mylist1.forEach(mycust -> System.out.println(mycust));
        System.out.println("-----");
    }
}
```



### // 3.Get Customers By City

```
List<Customer> mylist2 = custDAO.getCustomersByCity("Blore");
mylist2.forEach(mycust -> System.out.println(mycust));

System.out.println("Done!!!");
}
}
```

## 2)JLAppConfig.java

```
package com.jlcindia.springboot;

import java.util.Properties;
import javax.sql.DataSource;
import org.hibernate.SessionFactory;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.Bean;
import org.springframework.orm.hibernate5.HibernateTemplate;
import org.springframework.orm.hibernate5.HibernateTransactionManager;
import org.springframework.orm.hibernate5.LocalSessionFactoryBean;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
@SpringBootApplication
public class JLAppConfig {

    @Bean
    public LocalSessionFactoryBean getSessionFactory(DataSource mydataSource) {
        Properties myprops = new Properties();
        myprops.put("hibernate.show_sql","true");
        myprops.put("hibernate.dialect","org.hibernate.dialect.MySQL5Dialect");
        //myprops.put("hibernate.hbm2ddl.auto","update");

        LocalSessionFactoryBean mysessionFactory = new LocalSessionFactoryBean();

        mysessionFactory.setDataSource(mydataSource); //1
        mysessionFactory.setHibernateProperties(myprops); //2
        mysessionFactory.setPackagesToScan("com.jlcindia.springboot"); //3
        return mysessionFactory;
    }

    @Bean
    public HibernateTemplate getHT(SessionFactory mysessionFactory) {
        return new HibernateTemplate(mysessionFactory);
    }
}
```



@Bean

```
public HibernateTransactionManager getTxManager(SessionFactory sessionFactory) {  
    return new HibernateTransactionManager(sessionFactory);  
}  
}
```

### 3) Customer.java

```
package com.jlcindia.springboot;  
  
import javax.persistence.*;  
/*  
 * @Author : Srinivas Dande  
 * @Company: Java Learning Center  
 */  
@Entity  
@Table(name="mycustomers")  
public class Customer {  
    @Id  
    @GeneratedValue(strategy = GenerationType.IDENTITY)  
    @Column(name="cid")  
    private int cid;  
  
    @Column(name="cname")  
    private String cname;  
  
    @Column(name="email")  
    private String email;  
  
    @Column(name="phone")  
    private int phone;  
  
    @Column(name="city")  
    private String city;  
  
    public Customer() {}  
    public Customer(String cname, String email, int phone, String city) {  
        this.cname = cname;  
        this.email = email;  
        this.phone = phone;  
        this.city = city;  
    }  
  
    public Customer(int cid, String cname, String email, int phone, String city) {  
        super();  
        this.cid = cid;  
        this.cname = cname;
```



```
        this.email = email;
        this.phone = phone;
        this.city = city;
    }

    //Setters and Getters
    //toString()
}
```

#### 4) CustomerDAO.java

```
package com.jlcindia.springboot;

import java.util.List;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */
public interface CustomerDAO {
    public void addCustomer(Customer cust);
    public List<Customer> getAllCustomers();
    public List<Customer> getCustomersByCity(String city);
}
```

#### 5) CustomerDAOImpl.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.hibernate.criterion.DetachedCriteria;
import org.hibernate.criterion.Restrictions;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.orm.hibernate5.HibernateTemplate;
import org.springframework.stereotype.Repository;
import org.springframework.transaction.annotation.Transactional;
/*
 * @Author : Srinivas Dande
 * @Company: Java Learning Center
 */

@Repository
@Transactional
public class CustomerDAOImpl implements CustomerDAO {

    @Autowired
    HibernateTemplate htemp;
```



```
@Override
public void addCustomer(Customer cust) {
    htemp.save(cust);
}

@Override
public List<Customer> getAllCustomers() {

    List<Customer> mylist = htemp.loadAll(Customer.class);
    return mylist;

}

@Override
public List<Customer> getCustomersByCity(String city) {
    /*
    String HQL = "from Customer cust where cust.city=?";
    List<Customer> mylist =(List<Customer>) htemp.find(HQL, city);
    */

    DetachedCriteria dc=DetachedCriteria.forClass(Customer.class);
    dc=dc.add(Restrictions.eq("city",city));
    List<Customer> mylist = (List<Customer>) htemp.findByCriteria(dc);
    return mylist;
}

}
```

## **6)application.properties**

### **#Database Properties**

```
spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/myjlcdb5
spring.datasource.username=root
spring.datasource.password=srinivas
```

### **#Log Properties**

```
logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %m %n
```



### 7)pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo1-boot-hibernate</artifactId>
<version>1.0</version>
<name>demo1-boot-hibernate</name>
<description>This is Spring Boot and Hibernate Application</description>
<url>https://www.jlcindia.com</url>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.3.3.RELEASE</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>
</project>
```